Network Layer & Routing

OSI Layer 3 • IP Addressing • Routing Protocols • IPv4 vs IPv6

1. The Network Layer in the OSI Model

The Network Layer (Layer 3) is responsible for logical addressing, packet forwarding, and routing across interconnected networks. It provides the mechanisms needed to move data from a source host to a destination host across multiple intermediate nodes (routers), regardless of the underlying physical network technology.

Key responsibilities of Layer 3 include:

- Logical addressing — assigning IP addresses to identify hosts globally
- Routing — determining the best path for packets through the network
- Packet forwarding — moving packets hop-by-hop toward the destination
- Fragmentation & reassembly — splitting oversized packets for different link MTUs
- Error reporting — using ICMP to signal unreachable destinations or TTL expiry

OSI Layer Stack — The Network Layer highlighted:

OSI Layer	Protocols	PDU / Role
7 – Application	HTTP, FTP, DNS, SMTP	Application data
6 – Presentation	SSL/TLS, JPEG, ASCII	Encoding, encryption
5 – Session	NetBIOS, RPC	Session management
4 – Transport	TCP, UDP	Segments / datagrams
3 – Network ★	**IP, ICMP, OSPF, BGP**	**Packets**
2 – Data Link	Ethernet, Wi-Fi, PPP	Frames
1 – Physical	Cables, fiber, radio	Bits

2. IP Addressing

2.1 IPv4 Address Structure

An IPv4 address is a 32-bit number written as four decimal octets separated by dots (e.g. 192.168.10.1). It is divided into a network portion and a host portion, determined by a subnet mask.

> **Example**
> IP: 192.168.1.45   Subnet Mask: 255.255.255.0 (/24)Network: 192.168.1.0   Host: .45   Broadcast: 192.168.1.255Usable hosts: 254 (192.168.1.1 – 192.168.1.254)

2.2 IPv4 Address Classes

Class	Range	Default Mask	Network / Host	Usage
A	1.0.0.0 – 126.255.255.255	255.0.0.0 /8	8 / 24 bits	Large corps, ISPs
B	128.0.0.0 – 191.255.255.255	255.255.0.0 /16	16 / 16 bits	Mid-size orgs
C	192.0.0.0 – 223.255.255.255	255.255.255.0 /24	24 / 8 bits	Small networks
D	224.0.0.0 – 239.255.255.255	N/A	Multicast	OSPF, video streams
E	240.0.0.0 – 255.255.255.255	N/A	Reserved	Research / future

2.3 Private (RFC 1918) Address Ranges

- Class A: 10.0.0.0 – 10.255.255.255 /8 (16 million addresses)
- Class B: 172.16.0.0 – 172.31.255.255 /12 (1 million addresses)
- Class C: 192.168.0.0 – 192.168.255.255 /16 (65,536 addresses)

Private addresses are not routable on the public Internet. NAT (Network Address Translation) is used to map them to public IPs.

2.4 IPv4 Packet Header

Every IPv4 packet begins with a 20-byte minimum header containing fields that routers use to forward, fragment, and validate packets:

Field	Size	Description
Version	4 bits	IP version (4 or 6)
IHL	4 bits	Header length in 32-bit words
DSCP / ECN	8 bits	Quality of Service & congestion notification
Total Length	16 bits	Entire packet size (header + data), max 65,535 bytes
Identification	16 bits	Fragment group identifier
Flags / Offset	19 bits	Fragmentation control & fragment position
TTL	8 bits	Hop counter; decremented each router; drop at 0
Protocol	8 bits	Upper-layer protocol (6=TCP, 17=UDP, 1=ICMP)
Header Checksum	16 bits	Error detection for header only
Source IP	32 bits	Sender's IPv4 address

Field	Size	Description
Destination IP	32 bits	Recipient's IPv4 address
Options	Variable	Rarely used; timestamp, security, route recording

3. Routing

3.1 What is Routing?

Routing is the process of selecting paths through a network for traffic to travel from source to destination. Routers make forwarding decisions by examining each packet's destination IP address and consulting their routing table.

> **Key Concept — Routing vs Forwarding**
> Routing: the control-plane process of building and maintaining routing tables.Forwarding: the data-plane action of sending an incoming packet out the correct interface, guided by the routing table.

3.2 The Routing Table

Every router maintains a routing table — a list of known network destinations paired with the next-hop router and outgoing interface. The router performs longest-prefix matching: the most specific route (longest subnet mask) wins.

Sample Routing Table (C = Connected, S = Static, O = OSPF):

Destination	Next Hop	Interface	Src	AD	Notes
192.168.1.0/24	0.0.0.0	eth0	**C**	0	Directly connected LAN
10.0.0.0/8	192.168.1.1	eth0	**S**	1	Static route to HQ
172.16.0.0/12	10.10.10.1	eth1	**O**	110	Learned via OSPF
0.0.0.0/0	203.0.113.1	eth2	**S**	1	Default gateway (Internet)
203.0.113.0/30	0.0.0.0	eth2	**C**	0	ISP uplink (directly connected)

AD = Administrative Distance: a trustworthiness score (lower = more preferred). Connected = 0, Static = 1, OSPF = 110, RIP = 120, BGP = 20 (eBGP) / 200 (iBGP).

3.3 Types of Routing

Type	Description	Example
Static Routing	Manually configured routes. Simple, predictable, no overhead. Does not adapt to failures.	ip route 10.0.0.0 255.0.0.0 192.168.1.1
Dynamic Routing	Routes learned automatically via routing protocols. Adapts to topology changes.	OSPF, EIGRP, BGP announcing and withdrawing prefixes
Default Route	Catch-all route (0.0.0.0/0) for traffic with no more-specific match.	Gateway of last resort to the ISP
Policy Routing	Routes traffic based on criteria beyond destination IP (source, port, ToS).	Route VoIP traffic over low-latency link

4. Routing Protocols

Dynamic routing protocols automate the exchange of network topology or reachability information between routers, enabling them to build routing tables without manual configuration.

Protocol	Algorithm	Type	Metric	Use Case
RIP	Distance Vector	Interior (IGP)	Hop count (max 15)	Small / legacy LANs
OSPF	Link State	Interior (IGP)	Cost (bandwidth-based)	Enterprise networks
EIGRP	Hybrid	Interior (IGP)	Bandwidth + delay	Cisco environments
IS-IS	Link State	Interior (IGP)	Cost	ISP backbones
BGP	Path Vector	Exterior (EGP)	AS path + policies	Internet (between ASes)

4.1 Interior vs Exterior Gateway Protocols

- IGP (Interior Gateway Protocol) — operates within a single Autonomous System (AS). Examples: OSPF, EIGRP, RIP, IS-IS.
- EGP (Exterior Gateway Protocol) — exchanges routes between different ASes. Today only BGP is used.
- Autonomous System (AS) — a collection of IP networks under a single routing policy, identified by an AS Number (ASN).

4.2 Routing Algorithm Types

Routing protocols use three fundamental algorithmic approaches to discover and select paths:

Algorithm	Used In	Complexity	How It Works	Strength
Dijkstra's (SPF)	Link State (OSPF, IS-IS)	O((V+E) log V)	Complete topology map → computes shortest tree	Fast convergence, scales well
Bellman-Ford	Distance Vector (RIP)	O(V × E)	Iterative updates from neighbours	Handles negative weights; slow convergence
DUAL	Hybrid (EIGRP)	O(E)	Feasibility condition for loop-free backup paths	Fast failover, low overhead
BGP Decision	Path Vector (BGP)	Policy-driven	Weight → Local Pref → AS-path → MED → ...	Full Internet policy control

5. Key Protocols in Depth

5.1 OSPF (Open Shortest Path First)

OSPF is the most widely deployed IGP in enterprise and service provider networks. It is a link-state protocol — each router builds a complete map (LSDB) of the network and runs Dijkstra's SPF algorithm to compute loop-free shortest paths.

- Protocol: IP protocol 89 (not TCP/UDP)
- Metric: Cost = 10^8 / interface bandwidth (lower cost = faster link)
- Hello interval: 10 seconds (LAN), 30 seconds (NBMA)
- Dead interval: 4× Hello (40s / 120s)
- DR/BDR election: Reduces flooding on broadcast segments
- Areas: Hierarchical design — backbone (Area 0) plus stub/NSSA areas

> **OSPF Neighbour States**
> Down → Init → 2-Way → ExStart → Exchange → Loading → FullRouters must reach Full state before exchanging LSAs and computing routes.

5.2 BGP (Border Gateway Protocol)

BGP is the protocol that runs the Internet. It is an Exterior Gateway Protocol exchanging reachability information between Autonomous Systems. BGP is a path-vector protocol — it carries the full AS-path to each destination, which prevents routing loops.

- Transport: TCP port 179 (reliable delivery)
- iBGP: BGP sessions within the same AS (full mesh or route reflectors)
- eBGP: BGP sessions between different ASes
- Best-path selection: Weight → Local Preference → Locally originated → AS-path length → Origin → MED → eBGP > iBGP → IGP metric → Router-ID
- Convergence: Slow by design — stability and policy control are prioritised over speed

6. IPv4 vs IPv6

IPv4 address exhaustion (the last blocks were allocated in 2011) drove the development of IPv6, standardised in RFC 2460. IPv6 provides a vastly larger address space and simplifies header processing.

Feature	IPv4	IPv6
Address size	32-bit (4 bytes)	128-bit (16 bytes)
Notation	Dotted decimal (x.x.x.x)	Colon-hex (x:x:x:x:x:x:x:x)
Address space	~4.3 billion	~3.4 × 10^{38}
Header size	20–60 bytes (variable)	40 bytes (fixed)
Fragmentation	Routers & end hosts	End hosts only
Checksum	Yes (per-hop)	No (delegated to L4)
NAT required	Yes (address exhaustion)	No (vast space)
Auto-config	DHCP	SLAAC + DHCPv6
Security	IPSec optional	IPSec built-in
Broadcast	Yes	No (uses multicast/anycast)

6.1 IPv6 Address Types

- Unicast — one-to-one. Global unicast (2000::/3) replaces public IPv4; link-local (FE80::/10) mandatory on every interface.
- Multicast — one-to-many. FF00::/8. Replaces broadcast entirely.
- Anycast — one-to-nearest. Same address assigned to multiple interfaces; routing delivers to the closest one.

7. CIDR & NAT

7.1 CIDR (Classless Inter-Domain Routing)

CIDR, introduced in 1993, replaced the rigid class-based scheme with variable-length subnet masks (VLSM), allowing address space to be allocated in any power-of-two block size.

- Notation: 192.168.10.0/26 → 26-bit mask → 64 addresses, 62 usable hosts
- Route summarisation: Multiple contiguous subnets advertised as one aggregate, reducing routing table size
- Supernetting: Combining multiple networks into a larger block for aggregation

7.2 NAT (Network Address Translation)

NAT maps private IP addresses to one or more public IP addresses, allowing many hosts to share a single public IP.

- Static NAT — one private IP ↔ one public IP (1:1 mapping)
- Dynamic NAT — pool of public IPs shared by internal hosts
- PAT / NAT Overload — many private IPs map to one public IP, differentiated by port numbers (most common home/office setup)

8. Summary

The Network Layer is the foundation of internetworking. It provides the logical addressing and routing machinery that allows packets to traverse heterogeneous networks from any source to any destination on Earth.

- IP (v4 and v6) provides the universal addressing scheme used across all networks.
- Routing protocols (RIP, OSPF, EIGRP, BGP) automate path discovery and selection.
- Dijkstra, Bellman-Ford, and path-vector algorithms underpin routing decisions.
- CIDR and VLSM enable efficient address allocation and route summarisation.
- NAT bridges the gap between the exhausted IPv4 space and the Internet.
- IPv6 is the long-term solution: 128-bit addressing, simplified headers, built-in security.

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